UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 1

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Motivating examples for optimization.

Example 1 (Oil production management)**.** An oil refinary produces two products: jet fuel and gasoline.

- Profit for refinary is \$0.10 per barrel for jet fuel.
- Profit for refinary is \$0.20 per barrel for gasoline.
- Only 10,000 barrels are available for processing.
- Refinary has a contract to product at least 1,000 barrels of jet fuel and 2,000 barrels of gasoline.
- Both products are shipped in trucks, and the delivery capacity of the truck fleet is 180,000 barrel-miles.
- Jet fuel is delivered to an airfield 10 miles from the refinary, while gasoline is transported 30 miles to the distributor.

Question: How much of each product should be produced for maximum profit?
Answer:

1. Denote x_1 as the quantity (in barrels) of jet fuel, and x_2 as the quantity (in barrels) of gasoline.

   x_1 and x_2 are unknown variables.

2. The profit is $0.1x_1 + 0.2x_2$.

3. Constraints/conditions to be met:

   (a) $x_1 + x_2 \leq 10,000$ (limit on barrels).
   (b) $x_1 \geq 1,000$ (minimum jet fuel).
   (c) $x_2 \geq 2,000$ (minimum gasoline).
   (d) $10x_1 + 30x_2 \leq 180,000$ (fleet capacity).

Optimization:

$$\begin{aligned} \underset{x_1,x_2}{\text{maximize}} \quad & 0.1x_1 + 0.2x_2, \\ \text{subject to} \quad & \text{(a),(b),(c),(d)}. \end{aligned}$$

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Example 2 (Knapsack problem)**.** Consider a set of items $1, 2, \ldots, n$.

- Item $i \in \{1, 2, \ldots, n\}$ has weight w_i and value (importance) v_i.
- The bag can carry at most weight of w.

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Problem: Select a subset of items to put in the bag to maximize the total value.
Solution:

1. Unknown/variable: x_i for item $i \in \{1, 2, \ldots, n\}$.
   - x_i is set to 1 if item i is selected.
   - x_i is set to 0 if item i is not selected.
2. Objective: $\underbrace{\underbrace{v_1 x_1}_{\text{individual value}} + v_2 x_2 + \cdots + v_n x_n}_{\text{total value}}$.
3. Constraint: $\underbrace{\underbrace{w_1 x_1}_{\text{individual weight}} + w_2 x_2 + \cdots + w_n x_n}_{\text{total weight}} \leq w$.

Optimization:

$$\begin{aligned} \underset{x_i\text{'s}}{\text{maximize}} \quad & \sum_{i=1}^{n} v_i x_i, \\ \text{subject to} \quad & \sum_{i=1}^{n} w_i x_i \leq w, \\ & \underbrace{\boxed{x_i \in \{0,1\}}}_{x_i(x_i-1)=0}, \ i = 1, \ldots, n. \end{aligned}$$

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Standard form of optimization.

- Put all variables/unknowns into a vector $\boldsymbol{x} \in \mathbb{R}^n$. Express the optimization as

$$\begin{aligned} \min_{\boldsymbol{x} \in \mathbb{R}^n} \quad & f_0(\boldsymbol{x}), \\ \text{subject to} \quad & f_1(\boldsymbol{x}) \leq 0, \\ & \vdots \\ & f_m(\boldsymbol{x}) \leq 0. \end{aligned}$$

  Here min stands for minimize, and "subject to" is often shorten as s.t.

- Objective: $f_0(\boldsymbol{x})\colon \mathbb{R}^n \to \mathbb{R}$.
- There are m constraints; each $f_i(\boldsymbol{x})$ is from $\mathbb{R}^n$ to $\mathbb{R}$. So, $f_0, f_1, \ldots, f_m$ are all *scalars*.
- Right-hand sides of constraints are all zero.

Note: This is just one of the standard forms of optimization. Other standard forms exist for different applications.

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Example 3. Consider the following optimization problem:

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & x_1^2 + 5x_2^2 - 10x_1x_2, \\ \text{s.t.} \quad & 2 \le x_1 \le 5, \\ & x_1 + x_2 \ge 10. \end{aligned}$$

Put this into standard form:

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & \underbrace{\boxed{x_1^2 + 5x_2^2 - 10x_1x_2}}_{f_0(\boldsymbol{x})}, \\ \text{s.t.} \quad & \underbrace{\boxed{2 - x_1}}_{f_1(\boldsymbol{x})} \le 0, \\ & \underbrace{\boxed{x_1 - 5}}_{f_2(\boldsymbol{x})} \le 0, \\ & \underbrace{\boxed{10 - x_1 - x_2}}_{f_3(\boldsymbol{x})} \le 0. \end{aligned}$$

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- What if the objective is a maximization?

$$\max f(\boldsymbol{x}) \xrightarrow[\text{convert}]{} \min -f(\boldsymbol{x}).$$

- What if there are equality constraints?

$$h(\boldsymbol{x}) = 0 \xrightarrow[\text{convert}]{} \begin{cases} h(\boldsymbol{x}) \le 0, \\ -h(\boldsymbol{x}) \le 0. \end{cases}$$

So, each equality constraint can be replaced with two inequality constraints.

Definition 1 (Feasible solution)**.** We call $\boldsymbol{y} \in \mathbb{R}^n$ a *feasible solution/point* if it satisfies all constraints:

$$f_1(\boldsymbol{y}) \le 0,\ f_2(\boldsymbol{y}) \le 0,\ \ldots,\ f_m(\boldsymbol{y}) \le 0.$$

- A point that is not feasible is called an *infeasible point.*
- *Feasible set*: the set of all feasible solutions

$$\mathcal{X} = \{\boldsymbol{x} \in \mathbb{R}^n \mid f_1(\boldsymbol{x}) \le 0,\ f_2(\boldsymbol{x}) \le 0,\ \ldots,\ f_m(\boldsymbol{x}) \le 0\}.$$

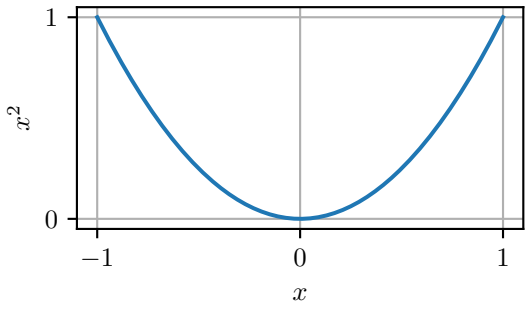
- A point $\boldsymbol{x}^* \in \mathbb{R}^n$ is called a *global solution* or a *global minimum* if

$$f_0(\boldsymbol{x}^*) \le f_0(\boldsymbol{x}) \quad \forall \boldsymbol{x} \in \mathcal{X}.$$

In other words, $\boldsymbol{x}^*$ returns the lowest objective value among all feasible points.

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Example 4. Minimize x^2 over the interval $[-1, 1]$:



- Point $x = \pm 5$: infeasible.
- Point $x = 0.5$: feasible.
- Point $x = 0$: global minimum.

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Definition 2 (Local minimum)**.** A point $\boldsymbol{x}^*$ is called a *local minimum* or *local solution* is there exists a positive number $r > 0$ such that

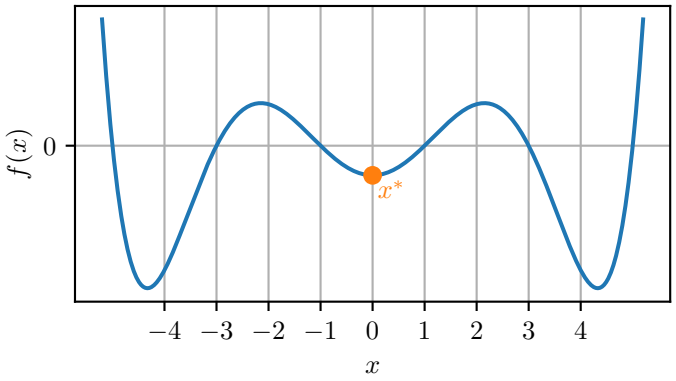
$$f_0(\boldsymbol{x}^*) \leq f_0(\boldsymbol{x}), \quad \forall \boldsymbol{x} \in \mathcal{X} \text{ and } \|\boldsymbol{x} - \boldsymbol{x}^*\| \leq r.$$

Here, $\|\cdot\|$ is the 2-norm:

$$\|\boldsymbol{x} - \boldsymbol{x}^*\| \leq r \iff \sqrt{(x_1 - x_1^*)^2 + \cdots + (x_n - x_n^*)^2} \leq r.$$

Therefore, $\boldsymbol{x}^*$ should be the best point in a ball around it of radius r.

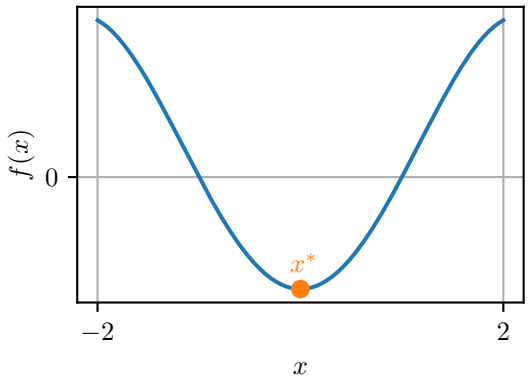
Example 5. Minimize $f(x)$ over $\mathbb{R}$:



Here, $x^* = 0$ is a local minimum, and r could be picked as any number less than 3 (or slightly more).

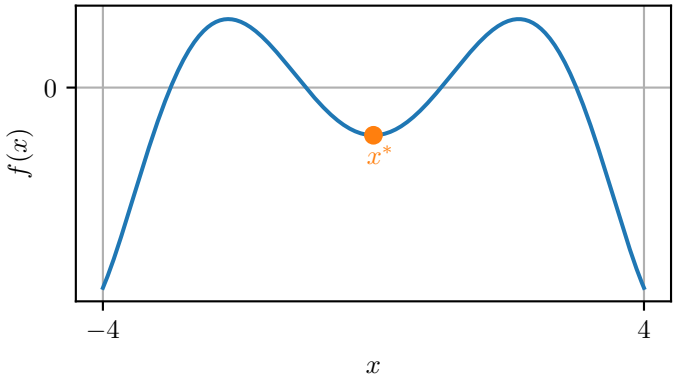
- When $r = 2$, this is the "ball":

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Here, $x^* = 0$ gives the lowest value in the "ball" $\|x - x^*\| \leq r$.

- When $r = 4$, $x^* = 0$ does not give the lowest value in the "ball":



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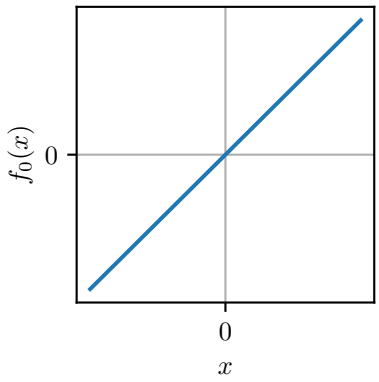
Definition 2 says some value of r should exist for a point to be a local minimum, and it doesn't need to hold for all values of r.

For example, in Example 5, $r = 2$ works while $r = 4$ doesn't.

Usually r is very small. If $r = +\infty$, then a local minimum becomes a global minimum.
When we solve an optimization problem, we obtain $\boldsymbol{x}^*$ and $f_0(\boldsymbol{x}^*)$. Here, $f_0(\boldsymbol{x}^*)$ is called an optimal objective value, and $\boldsymbol{x}^*$ is called a minimizer in some textbooks.
Sometimes $\boldsymbol{x}^*$ does not exist:

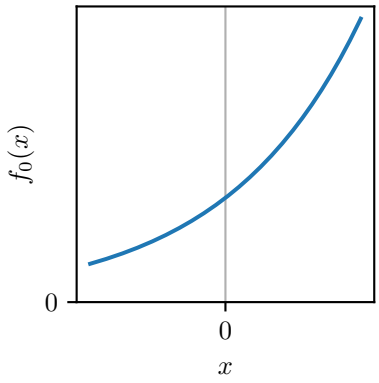
1. $\min x$:

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$x^* = -\infty$ is not a number.

2. $\min \mathrm{e}^x$:



$x^* = -\infty$ is not a valid choice.

In case 1, the optimal objective value is $-\infty$; in case 2, the optimal objective value is 0 (it is the limit of the objective).

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